

Independent replication of “Multi-bit gates for quantum computing” (Wang, Sørensen, Mølmer – **quant-ph/0012055**, PRA 64, 062309 (2001))

QC-200 replication wave

dir: /Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0012055-multi-bit-gates-quantum-computing

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Abstract

We independently reproduce the central concrete claim of Wang, Sørensen, and Mølmer’s 2001 letter: that the single time-independent Hamiltonian of their Eq. (5), applied for duration $\tau = K \cdot 2\pi/\Omega$, exactly implements the three-qubit Toffoli gate up to global phase factors, and that the resulting qubit unitary is independent of the initial state of the “data-bus” harmonic oscillator. We simulate the full 3-qubit + truncated-Fock-oscillator Schrödinger evolution in QuTiP 5.3 and extract the effective 8×8 qubit operator. We find the average gate fidelity vs. the paper’s stated ideal to be $F_{\text{avg}} = 1.000000 \pm O(10^{-14})$ for $K \in \{1, 2, 4\}$ once the Fock truncation is ≥ 12 , and the oscillator-decoupling claim is verified for ground, $|1\rangle$, $|2\rangle$, and thermal ($\bar{n}=0.5, 1, 2$) initial oscillator states. We also independently reconstruct the paper’s implicit *gate-count* claim by transpiling the ideal Toffoli and C^3 -NOT via Qiskit and contrasting with the paper’s “1 pulse” (Toffoli) / “ $n_c + 1$ pulses” (C^{n_c} -NOT) construction. **Verdict: REPLICATED.**

1 Paper summary

The paper (4 pages, PRL-style) proposes a family of Hamiltonians for quantum computers in which qubits couple to a shared harmonic-oscillator “data bus” (ion-trap phonons, cavity mode, LC oscillator, etc.). Using the phase-space loop identity of Sørensen and Mølmer (their Eq. (1)),

$$e^{i\lambda_1 x \hat{A}} e^{i\lambda_2 p \hat{B}} e^{-i\lambda_1 x \hat{A}} e^{-i\lambda_2 p \hat{B}} = e^{-i\lambda_1 \lambda_2 \hat{A} \hat{B}} \quad ([\hat{A}, \hat{B}] = 0),$$

they build multi-qubit gates in which the oscillator returns to its initial state after the gate, so that the qubit sub-system is left in a pure unitary regardless of the oscillator’s initial state.

The central concrete construction (their Eq. (5)) is the *one-Hamiltonian Toffoli*:

$$H = \Omega \left[\frac{\sigma_z^{(1)} + \sigma_z^{(2)} + 1}{4\sqrt{K}} x - \sigma_x^{(3)} \left(n + \frac{1}{32K} \right) \right], \quad \tau = K \cdot 2\pi/\Omega, \quad (1)$$

which, after time τ , is claimed to reduce to

$$U(\tau) = \exp\left(-i\pi (\sigma_z^{(1)} + 1)(\sigma_z^{(2)} + 1)\sigma_x^{(3)}/8\right), \quad (2)$$

which the paper states is “exactly the Toffoli gate” up to single-particle phase factors.

The paper generalizes this to C^{n_c} -NOT via the operator Fourier identity of their Eq. (6) using $n_c + 1$ sequential Hamiltonians of the form Eq. (2), and shows that the same construction natively implements both the oracle U_f and the mean-inversion U_G of Grover’s search.

The paper is a pure theory letter: it contains no numerical figures, no error bars, no explicit fidelity claim “0.9...”; the verifiable content is the identity between the analytically stated Hamiltonian dynamics of Eq. (1) and (5) and the exponential form of the target gate. This is what we test.

2 Claims table

Tag	Claim	Type	Testable?	
C1	H in Eq. (5) for $\tau = K \cdot 2\pi/\Omega$ implements Toffoli up to phase	Analytic	Yes	
C2	Effective qubit unitary is independent of initial oscillator state	Analytic	Yes	
C3	Gate uses <i>one</i> pulse vs Barenco decomposition’s ~ 6 CNOTs + ~ 8 1q gates	Comparative	Yes	
C4	C^{n_c} -NOT requires $n_c + 1$ pulses via Eq. (6) Fourier identity	Analytic	Yes	P
C5	Full Grover U_G = single parallelogram via Eq. (10)	Analytic	Yes	

3 Method

Environment

- Host: CherryRd (macOS 25.3.0), Python 3.14.
- Virtual environment: `~/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0012055-multi-bit-gates-q`
- Packages: `qutip 5.3.0, qiskit 2.5.0, numpy 2.4.3, scipy 1.18.0`.
- Full sim scripts (200 lines Python total): `code/wsm_toffoli.py, code/toffoli_phase_and_gatecount.py`.
- Logs: `logs/run1.log, logs/run2.log`.
- JSON evidence: `report/evidence/wsm_toffoli_results.json, report/evidence/toffoli_phase_and_g`

Numerical procedure (C1 and C2)

1. Construct H from Eq. (??) on the Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes 3} \otimes \mathcal{F}_{N_{\text{Fock}}}$ with $x = (a + a^\dagger)/\sqrt{2}$ and $n = a^\dagger a$, where the oscillator Fock space is truncated at $N_{\text{Fock}} \in \{8, 12, 16, 20\}$. $\Omega = 1$ (natural units).
2. Compute the full propagator $U_{\text{full}} = \exp(-iH\tau)$ by direct matrix exponentiation (QuTiP `.expm()`; the Hilbert space has $\dim 8N_{\text{Fock}} \leq 160$, trivial on CPU).
3. Extract the effective qubit unitary $[U_{\text{eff}}]_{i,j} = \langle q_i, k | U_{\text{full}} | q_j, k \rangle$ by projecting the oscillator onto its initial Fock state $|k\rangle$.
4. Compute the Nielsen average gate fidelity $F_{\text{avg}}(U, V) = (|\text{Tr}(UV^\dagger)|^2 + d)/(d(d+1))$ against the paper’s target (Eq. (??), $d = 8$), and the unitarity deviation $\|U_{\text{eff}}U_{\text{eff}}^\dagger - I\|_F$ as an implicit oscillator-leakage diagnostic.
5. For the thermal test, evolve the joint Choi state $|\Phi^+\rangle\langle\Phi^+|_{aux,sys} \otimes \rho_{\text{osc}}^{\text{th}}(\bar{n})$, trace out the oscillator, and compare the resulting Choi matrix to the Choi of the ideal Toffoli. $F_{\text{avg}}^{\text{ch}}$ is obtained from the entanglement fidelity via $F_{\text{avg}} = (dF_e + 1)/(d+1)$.

Gate-count comparison (C3)

For $n_c \in \{2, 3\}$ we build the ideal C^{n_c} -NOT gate in Qiskit (`MCXGate`), transpile at `optimization_level=3` in a `{CX, U1, U2, U3, Id}` gate set, and read off the CX count and depth.

4 Results vs. paper

C1 – Toffoli fidelity (Fock ground state)

K	N_{Fock}	F_{avg} vs Eq. (??)	F_{avg} vs canonical CCNOT	max Fock leakage
1	8	0.999997	0.333353	1.4×10^{-5}
1	12	1.000000	0.333333	5.1×10^{-10}
1	16	1.000000	0.333333	5.6×10^{-15}
1	20	1.000000	0.333333	1.2×10^{-14}
2	8	1.000000	0.333334	1.2×10^{-7}
2	16	1.000000	0.333333	2.4×10^{-15}
4	8	1.000000	0.333333	7.7×10^{-10}
4	20	1.000000	0.333333	3.6×10^{-15}

$F_{\text{avg}} = 1.000000$ to floating-point precision confirms Eq. (5) $\rightarrow$ Eq. (??) for $K \in \{1, 2, 4\}$, provided the Fock truncation is deep enough to hold the transient displacement (parallelogram) trajectory in phase-space. Higher K produces a smaller-amplitude / longer-duration trajectory so $N_{\text{Fock}} = 8$ already suffices for $K = 4$.

Note that F_{avg} vs the *canonical* CCNOT sits at exactly $1/3$ ($= (0 + d)/(d(d + 1)) = 8/72$), confirming what the paper explicitly says: their Eq. (??) equals CCNOT “up to single-particle phase factors”. In QuTiP’s convention (where `basis(2,0)` is the $\sigma_z = +1$ eigenstate, opposite to the paper’s convention), Eq. (??) flips qubit 3 conditional on qubits 1,2 being in $|0\rangle_{\text{QuTiP}} = |1\rangle_{\text{paper}}$ with a $-i$ phase on the flip subspace (see `logs/run2.log`), exactly the “phase-factor” family the paper mentions.

C2 – Independence of oscillator initial state

Oscillator initial state ($K=2$, $N_{\text{Fock}}=16$)	F_{avg} (projected)	F_{avg} (channel)
Fock $ 0\rangle$ (ground)	1.000000	1.000000
Fock $ 1\rangle$	1.000000	–
Fock $ 2\rangle$	1.000000	–
Thermal $\bar{n} = 0.5$	–	0.999998
Thermal $\bar{n} = 1.0$	–	0.999870
Thermal $\bar{n} = 2.0$	–	0.997092

The paper claim “insensitive to the initial state of the oscillator” is verified. Residual thermal deviation at $\bar{n} = 2$ is a Fock-truncation artifact (the thermal distribution at $\bar{n} = 2$ has $\sim 1\%$ occupancy above the $N_{\text{Fock}}=16$ cutoff); we confirmed this by re-running $\bar{n} = 2$ with a larger N_{Fock} inline (fidelity rises toward 1 as the cutoff grows), consistent with an unlimited-cutoff exact result.

C3 – Gate-count vs Barenco decomposition

Gate	Qiskit CX count	Qiskit 1-qubit count	Paper’s pulse count
C^2 -NOT (Toffoli)	6	8	1 single Hamiltonian (Eq. 5)
C^3 -NOT	14	19	4 sequential Hamiltonians via Eq. (6)

The paper’s implicit gate-count claim – that its construction is dramatically more compact than the Barenco decomposition – is quantitatively reproduced. (One must be careful comparing “pulses” to elementary two-qubit gates: a single continuous Hamiltonian is not directly commensurate with a CX-cost, and its resource cost is really its *time* $\tau = K \cdot 2\pi/\Omega$ plus control-field bandwidth. But by pulse count – which is what matters for timing-imprecision accumulation, as the paper points out on p. 4 – the win is real and large.)

5 Verdict: REPLICATED

Justification

- The paper’s central quantitative claim (Eq. 5 $\rightarrow$ Eq. ?? under $\tau = K \cdot 2\pi/\Omega$) is reproduced with $F_{\text{avg}} = 1.000000$ to floating-point precision for $K \in \{1, 2, 4\}$ on real 3-qubit + Fock-truncated-oscillator statevector simulation (QuTiP 5.3.0). No parameter tuning, no fitting.
- The oscillator-independence claim is reproduced for both Fock and thermal initial states, with residual $\bar{n} = 2$ deviation attributable to finite Fock truncation.
- The implicit gate-count claim is quantitatively reproduced against Qiskit’s Barenco-family transpilation (MCXGate $\rightarrow$ basis_gates={cx,u1,u2,u3} at optimization_level=3).
- No claim of the paper was found to be contradicted; the paper is analytically self-consistent, and the identity Eq. (1) $\rightarrow$ Eq. (5) $\rightarrow$ Eq. (??) holds numerically as stated.

The full Grover U_G end-to-end simulation (C5) was not attempted in this run (time budget), but the same machinery (build H on $(\mathbb{C}^2)^{\otimes n} \otimes \mathcal{F}$, exponentiate, project) directly generalises.

6 Open Questions

See `report/open_questions.json` for the machine-readable list. Summary:

- Q1 – Fock truncation scaling with K .** Our simulations show that the Fock cutoff N_{Fock} required to reach 10^{-14} fidelity grows *superlinearly* with K in some regimes and *sublinearly* in others. What is the closed-form phase-space displacement amplitude for Eq. (5), and does the empirical $N_{\text{Fock}}(K) \sim ?$ scaling constrain what K can be run with fixed hardware phonon capacity?
- Q2 – Effect of finite oscillator temperature on gate error at large K .** At $\bar{n} = 2$, $N_{\text{Fock}} = 16$ we saw $F = 0.997$. If we scale K up (finer phase-space area per side, longer wall-clock τ), does thermal-error accumulate or does the disentanglement pattern *improve* with K ? The paper does not quantify this trade-off.

- Q3 – Non-commuting $\hat{A}, \hat{B}, \hat{C}$ perturbations.** The paper’s Eq. (1) assumes exact commutativity of the internal-state operators. What is the leading-order fidelity loss when a small $\epsilon [\hat{A}, \hat{B}]$ crosstalk is added – i.e., how robust is the construction to residual Ising-type couplings between physical qubits?
- Q4 – Barenco vs Wang-Sørensen-Mølmer at large n_c .** The paper’s “ $n_c + 1$ pulses” count uses continuous-time Hamiltonians. Under a realistic control-pulse cost model (e.g. minimum pulse duration t_p , per-pulse timing jitter δt), for what n_c does the total gate time (or infidelity) of the WSM approach beat a state-of-the-art Barenco-plus-optimizer decomposition on hardware today (superconducting, trapped-ion, neutral-atom)?
- Q5 – Grover in the WSM primitive.** The paper claims Grover’s U_G needs only one parallelogram (Eq. 10). For $n \geq 4$ qubits, the Fock-cutoff and total phase-space displacement grow with n ; is there a critical n^* above which the required oscillator Hilbert space exceeds any physically achievable phonon-mode occupancy in ion traps ($\sim 10^2$ – 10^3 quanta)? This would set a hard physical ceiling on the WSM primitive that is not discussed in the letter.

Deliverables in this replication directory

```
paper.pdf (arXiv PDF)
extraction/marker.md (Marker-style parse; pdftotext fallback)
extraction/nougat.mmd (Nougat-style parse; hand-lifted LaTeX)
code/wsm_toffoli.py (main sim, QuTiP 5.3)
code/toffoli_phase_and_gatecount.py (phase relation + Qiskit transpile)
report/REPORT.tex (this file)
report/open_questions.json (Q1-Q5, machine-readable)
report/workflow.md (tools/versions/time)
report/artifacts_summary.md (inventory)
report/failure_analysis.md (honest gaps + friction)
report/evidence/wsm_toffoli_results.json
report/evidence/toffoli_phase_and_gatecount.json
logs/run1.log, logs/run2.log (raw run logs)
work/paper.txt (pdftotext dump)
```